Linux Server Admin Interview Guide

Scenario-based answers for complete beginners

15 common interview questions explained from zero: what it is, a real-life story, the commands, and a short answer you can say in the interview.

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NOTE

A few questions in your list were written in a short form, so I had to interpret them: **NCL** (Q4), **NCP** (Q6) and **'How will come to know server command'** (Q14). I explain my interpretation inside each answer. If your interviewer meant something different, politely ask: *'Do you mean X?'* - that is perfectly acceptable in an interview.

How to use this guide: read the scenario first, then the simple explanation, then try the commands on a practice Linux machine (a free Ubuntu/CentOS virtual machine or WSL is enough). Finally, read the green box and practise saying it in your own words.

Start here: Linux in 5 minutes

If you know nothing yet, read this page twice. Everything else builds on it.

The basic words

Word	Simple meaning
Server	A powerful computer that runs applications and websites 24 hours a day for many users. It usually has no screen or keyboard; you control it from your laptop.
Linux	An operating system (like Windows), but most company servers use it. Versions are called distributions: RHEL, CentOS, Rocky, Ubuntu.
PuTTY	A small Windows program that connects to a Linux server over the network and gives you a black text window. This connection method is called **SSH** (port 22).
Terminal / CLI	The black text window where you type commands. CLI = Command Line Interface.
root	The all-powerful admin user. It can delete anything, so be careful. Normal admins use `sudo` to run single commands as root.
Process / Service	A running program. A service (like sshd, httpd) runs in the background and is managed with `systemctl`.

How a command looks

```
ls -l /etc
|   |  |
|   |  +-- argument  (what to work on: the /etc folder)
|   +----- option    (how to work: -l = long detailed list)
+--------- command   (what to do: list files)
```

Important folders (the map of Linux)

Folder	What lives there	Folder	What lives there
/	Top of everything (root)	/var/log	Log files (first place to look for errors)
/etc	Configuration files	/boot	Kernel and boot loader files
/home	Users' personal folders	/tmp	Temporary files
/dev	Devices (disks like /dev/sdb)	/data	Not standard - a folder admins create for application data

BE CAREFUL

Never run a command on a production (live) server if you do not understand it. Dangerous examples: `rm -rf` (deletes without asking), `mkfs` (erases a disk), `reboot` (restarts the server). Always take a backup or snapshot before changing anything, and read error messages slowly - they usually tell you the answer.

Q1. Port number for DNS

SCENARIO (a real-life situation)

A developer says: *'My application server can reach 8.8.8.8, but it cannot open google.com.'* Your senior says: *'Looks like a DNS problem. Check it.'*

Simple explanation

DNS is the phone book of the internet. Humans remember names (google.com) but computers talk using numbers (IP addresses). DNS converts the name into the number.

A **port** is a numbered door on a server. Each service uses its own door, so one server can run many services at once. **DNS uses port 53.**

Protocol	Used for
UDP 53	Normal DNS lookups (small and fast)
TCP 53	Large answers and **zone transfers** (copying DNS data from a primary DNS server to a secondary one)
TCP 853 / 443	Encrypted DNS (DNS over TLS / DNS over HTTPS) - a bonus point if you mention it

Commands to troubleshoot

```
cat /etc/resolv.conf        # which DNS server is this machine using?
nslookup google.com         # ask DNS: what is the IP of google.com?
dig google.com              # same, with more detail
dig @8.8.8.8 google.com     # ask one specific DNS server (proves port 53 is reachable)
ss -ulnp | grep :53         # on the DNS server: is a service listening on port 53?
```

If the name fails but the IP works, check: (1) the DNS server address in `/etc/resolv.conf`, (2) that the DNS server is up, (3) that the firewall allows port 53 on both UDP and TCP.

HOW TO ANSWER IN THE INTERVIEW (say this)

'DNS uses port 53. UDP 53 is used for normal queries and TCP 53 is used for zone transfers and large responses. If name resolution fails, I check /etc/resolv.conf, test with nslookup or dig against the DNS server, and confirm the firewall allows port 53.'

Other common ports to remember

Service	Port	Service	Port	Service	Port
SSH / SFTP	22	HTTP	80	MySQL	3306
FTP	21	HTTPS	443	PostgreSQL	5432
Telnet	23	NTP (time)	123	RDP (Windows)	3389
SMTP (mail)	25	LDAP	389	NFS	2049
DNS	53	SNMP	161	Syslog	514

Q2. How to check port numbers on servers

SCENARIO (a real-life situation)

The application team says: *'Our app runs on port 8080 but users cannot reach it.'* You must find out (a) is the app really listening on 8080 on the server, and (b) can another machine reach that port?

Part A - What is listening on THIS server?

```
ss -tulnp                  # modern command - lists all listening ports
netstat -tulnp             # older command, same idea (needs net-tools package)
ss -tulnp | grep 8080      # is anything listening on 8080?
lsof -i :8080              # which program owns port 8080?
```

Meaning of the letters: **`t`** = TCP, **`u`** = UDP, **`l`** = listening only, **`n`** = show numbers not names, **`p`** = show the program (PID/name).

How to read the output: a line like **`LISTEN 0 128 0.0.0.0:8080 users:(("java",pid=1234))`** means the Java program is listening on port 8080 on all network cards. If you see **`127.0.0.1:8080`** instead, the app only accepts connections from the same machine - that could be the cause of the problem.

Part B - Can another machine reach the port?

```
telnet 192.168.1.10 8080    # connects? then the port is open
nc -zv 192.168.1.10 8080    # nc = netcat, prints 'succeeded' if open
nmap -p 8080 192.168.1.10   # shows open / closed / filtered
```

Part C - Which port belongs to which service name?

```
grep -w ssh /etc/services   # the file that maps names to standard port numbers
firewall-cmd --list-all     # is the port allowed by the firewall? (RHEL/CentOS)
```

Result	Meaning
open / succeeded	Service is listening and the path is clear.
Connection refused	Server reachable, but nothing is listening on that port (service down).
timed out / filtered	A firewall or network device is blocking the traffic.

HOW TO ANSWER IN THE INTERVIEW (say this)

'On the server I use ss -tulnp (or netstat) to see listening ports and which process owns them. From another machine I test with telnet, nc or nmap. If it is refused, the service is down; if it times out, a firewall is likely blocking it.'

Q3. SFTP command line

SCENARIO (a real-life situation)

You must upload a configuration file from your laptop to a server and download a log file from the server. FTP is not allowed because it is not secure. Your manager says: *'Use SFTP.'*

Simple explanation

SFTP (SSH File Transfer Protocol) copies files between computers **securely**, using the SSH connection on **port 22**. Normal FTP sends passwords in plain text; SFTP encrypts everything.

Connect

```
sftp user1@192.168.1.10          # connect as user1 (asks for password)
sftp -P 2222 user1@192.168.1.10  # if SSH runs on a different port (capital -P)
```

A sample session (you type the lines after sftp>)

```
sftp> pwd                  # where am I on the SERVER (remote)?
sftp> lpwd                 # where am I on MY machine (local)?
sftp> ls                   # list files on the server
sftp> cd /tmp              # change folder on the server
sftp> lcd /home/me/files   # change folder on my machine
sftp> put config.txt       # UPLOAD  (my machine -> server)
sftp> get app.log          # DOWNLOAD (server -> my machine)
sftp> mput *.conf          # upload many files
sftp> mget *.log           # download many files
sftp> mkdir backup         # create a folder on the server
sftp> rm oldfile.txt       # delete a file on the server
sftp> exit                 # close the connection
```

Memory trick: put = push up to the server, **get** = grab down to me. Commands that start with **`l`** (lpwd, lcd, lls) work on your **l**ocal machine.

Related command: **`scp file.txt user1@192.168.1.10:/tmp/`** copies a file in one line (also over SSH).

HOW TO ANSWER IN THE INTERVIEW (say this)

'SFTP transfers files securely over SSH on port 22. I connect with sftp user@host, use put to upload, get to download, ls/cd/pwd on the remote side and lcd/lpwd for the local side.'

Q4. What is CLI (and NCL)?

NOTE

I am not fully sure what **'NCL'** means in your question. The most common interview pair is **CLI vs GUI** (Graphical User Interface), so I explain that. If the interviewer meant **ncli** - the command-line tool of Nutanix virtualization - the idea is the same: a CLI used to manage a system by typing commands. Ask the interviewer to clarify.

SCENARIO (a real-life situation)

A new colleague asks: *'Why do we type commands in a black window when Windows has nice buttons and menus?'*

Simple explanation

CLI (Command Line Interface) - you control the computer by typing text commands. **GUI (Graphical User Interface)** - you control it with a mouse, icons and windows.

Restaurant analogy: GUI is pointing at pictures on a menu. CLI is telling the chef exactly what you want in words - it takes some learning, but you can order very precisely, and you can order 100 dishes with one sentence.

	CLI	GUI
How	Type commands	Click icons and menus
Resources	Very light - ideal for servers	Heavy - needs more CPU and memory
Remote work	Works over slow links (SSH/PuTTY)	Needs a faster connection
Automation	Easy: put commands in scripts and cron jobs	Hard to automate
Learning	Harder at first	Easy for beginners

That is why servers use CLI: no screen is attached, it uses fewer resources, it works remotely, and tasks can be repeated by scripts on hundreds of servers.

HOW TO ANSWER IN THE INTERVIEW (say this)

'CLI is a text-based interface where we give commands to the operating system. Servers are managed through CLI because it is lightweight, works remotely over SSH, and can be automated with scripts. GUI is the graphical alternative, mostly used on desktops.'

Q5. Booting process

SCENARIO (a real-life situation)

After a power cut, a Linux server restarts but never reaches the login prompt. Your manager asks: *'Where exactly is it stuck?'* To answer, you must know the steps of booting.

Simple explanation - like starting a car

Step	What happens	Car analogy
1. Power on / BIOS or UEFI	Firmware does **POST** (Power-On Self Test): checks CPU, memory, keyboard. Then finds the boot disk.	Turn the key; dashboard lights check everything.
2. Boot loader (GRUB2)	Read from the first sector of the disk (MBR) or EFI partition. Shows the boot menu and loads the **kernel** and **initramfs** into memory. Config: `/boot/grub2/grub.cfg`	Pick the gear and start the engine.
3. Kernel	The core of Linux starts, detects hardware, and mounts the root filesystem (/) using temporary drivers from initramfs.	Engine runs.
4. systemd (PID 1)	The very first process. Reads the default **target**, mounts disks from `/etc/fstab`, starts network, ssh and other services.	Lights, AC, radio switch on.
5. Login prompt	The server is ready; you can login by console or SSH.	Drive away.

Useful commands

```
systemctl get-default       # multi-user.target (text) or graphical.target (GUI)
journalctl -b               # all messages from the current boot
dmesg | less                # kernel messages during boot
systemd-analyze blame       # which service made the boot slow?
```

Old runlevels you may be asked about: 0 = shutdown, 1 = single-user (repair), 3 = multi-user text mode, 5 = GUI, 6 = reboot.

Where is it stuck? (troubleshooting by stage)

Symptom	Stage	Typical fix
No display / 'no bootable device'	BIOS / disk	Check boot order, disk, cable or RAID.
'grub>' or 'grub rescue>' prompt	Boot loader	Boot from rescue media and reinstall or repair GRUB.
Kernel panic	Kernel / initramfs	Boot an older kernel from the GRUB menu; rebuild initramfs.
'Welcome to emergency mode'	systemd / fstab	Usually a wrong or missing disk in /etc/fstab; fix and reboot.

HOW TO ANSWER IN THE INTERVIEW (say this)

'Power on, BIOS/UEFI runs POST and finds the boot device, GRUB loads the kernel and initramfs, the kernel mounts the root filesystem and starts systemd as PID 1, systemd starts services according to the default target, and finally the login prompt appears.'

Q6. What is NCP / NFS? What if df -h hangs?

NOTE

I read **'ncp'** as **NFS** (Network File System), because the classic reason for `df -h` hanging is an NFS mount that stopped responding. There is also an old protocol called NCP (NetWare Core Protocol) that does a similar 'shared folder' job, but it is rarely used today. Confirm with the interviewer if unsure.

SCENARIO (a real-life situation)

You type `df -h` to check disk space and the screen just freezes - no output, no prompt. Other admins are waiting for the disk report.

Simple explanation

NFS lets one server use a folder that physically lives on another server, as if it were local - like a shared network drive. Example: `/mnt/share` on your server actually lives on storage server 10.0.0.50.

Why does df -h hang? `df` asks every mounted filesystem for its size. If an NFS server is down or the network is broken, the question never gets an answer and `df` waits forever.

Troubleshooting steps

- **Stay calm.** Press Ctrl+C. Open a second PuTTY session (the first one may stay stuck).
- **Check local disks only** - proves the problem is a network mount.

```
df -hl                      # -l = local filesystems only. Does this work?
timeout 10 df -h            # gives up after 10 seconds instead of hanging forever
```

- **Find the NFS mounts.**

```
mount | grep nfs
grep nfs /proc/mounts       # works even if some commands hang
```

- **Check the NFS server and network.**

```
ping 10.0.0.50              # is the NFS server alive?
nc -zv 10.0.0.50 2049       # is the NFS port reachable?
showmount -e 10.0.0.50      # does the server still export the folder?
dmesg | tail                # look for 'nfs: server 10.0.0.50 not responding'
```

- **Look for stuck processes** (state D = waiting for I/O).

```
ps -eo pid,stat,cmd | awk '$2 ~ /^D/'
```

- **Fix.** Best fix: restore the NFS server or network. If that is not possible, remove the stale mount:

```
umount -f /mnt/share        # force unmount
umount -l /mnt/share        # lazy unmount - detach now, clean up later
mount -a                    # remount everything from /etc/fstab once the server is back
```

BE CAREFUL

Do not randomly kill processes or reboot. Forced unmount may lose unsaved writes to that share. Inform the application owner before forcing anything on production.

HOW TO ANSWER IN THE INTERVIEW (say this)

'NFS shares a remote folder over the network. df -h hangs when an NFS server is unreachable. I use df -hl or timeout to confirm, find the NFS mounts with mount | grep nfs, check ping, port 2049 and showmount, look at dmesg, then fix the server or network, or use umount -f / -l and remount.'

Q7. I need yesterday's load on servers

SCENARIO (a real-life situation)

Users complained: *'The application was very slow yesterday afternoon.'* The manager asks for yesterday's CPU load. You cannot just run `uptime` - that only shows **now**.

Simple explanation

Load average = the average number of tasks waiting for or using the CPU. `uptime` shows three numbers: the average over the last **1, 5 and 15 minutes**. Compare it with the number of CPU cores (`nproc`). On a 4-core server, a load of 4 means fully busy; load 12 means heavily overloaded.

To see **history**, you need the tool **sar** from the **sysstat** package. It records performance data every 10 minutes into one file per day.

Commands

```
nproc                                              # number of CPU cores
ls /var/log/sa/                                    # daily files: sa01, sa02 ... (RHEL/CentOS)
                                                   # on Ubuntu: /var/log/sysstat/

sar -q -f /var/log/sa/sa$(date -d yesterday +%d)   # LOAD average for yesterday
sar -u -f /var/log/sa/sa$(date -d yesterday +%d)   # CPU usage for yesterday
sar -r -f /var/log/sa/sa$(date -d yesterday +%d)   # memory usage for yesterday
sar -q -s 13:00:00 -e 17:00:00 -f /var/log/sa/sa$(date -d yesterday +%d)
                                                   # only the afternoon window
```

In the `sar -q` output, look at the columns **ldavg-1, ldavg-5, ldavg-15** and **runq-sz** (tasks waiting to run). Find the time when the numbers jumped, then ask: which job ran then?

BE CAREFUL

If sysstat is not installed, yesterday's data does **not exist** - nobody can recover it from the server. Install and enable it for the future: `yum install sysstat` and `systemctl enable --now sysstat`. Also check whether the company has a monitoring tool (Zabbix, Nagios, Grafana, Dynatrace, CloudWatch) that stores history.

HOW TO ANSWER IN THE INTERVIEW (say this)

'uptime shows only the current load. For yesterday I use sar from the sysstat package, reading the previous day's file under /var/log/sa with sar -q for load, -u for CPU and -r for memory. If sysstat is not enabled, I check the monitoring tool, and I enable sysstat so history is available next time.'

Q8. How will you check packages in servers?

SCENARIO (a real-life situation)

The security team says: *'A serious bug was found in OpenSSL. Tell us which version is installed on our servers.'* You must check installed packages.

Simple explanation

A **package** is an installable software bundle (like an .exe installer on Windows). The package manager keeps a list of what is installed. The commands depend on the Linux family:

Task	RHEL / CentOS / Rocky	Ubuntu / Debian
List all installed	`rpm -qa` or `yum list installed`	`dpkg -l` or `apt list --installed`
Is X installed?	`rpm -q openssl`	`dpkg -l | grep openssl`
Details / version	`rpm -qi openssl`	`apt show openssl`
Files of a package	`rpm -ql openssl`	`dpkg -L openssl`
Which package owns a file?	`rpm -qf /usr/bin/ssh`	`dpkg -S /usr/bin/ssh`
Updates available?	`yum check-update`	`apt list --upgradable`
History of installs	`yum history`	see /var/log/apt/history.log

Checking many servers at once

```
for s in $(cat servers.txt); do
    echo "== $s =="; ssh $s "rpm -q openssl"
done
```

Larger companies use tools such as Ansible, Red Hat Satellite or a CMDB report for the same job.

HOW TO ANSWER IN THE INTERVIEW (say this)

'On RHEL-based systems I use rpm -qa, rpm -q package-name or yum list installed; on Ubuntu I use dpkg -l or apt list --installed. rpm -qi gives details and rpm -qf shows which package owns a file. For many servers I use a loop over SSH or Ansible.'

Q9. If you cannot login through PuTTY, how will you troubleshoot?

SCENARIO (a real-life situation)

Monday morning: you open PuTTY to a server you used on Friday, and it does not let you in. The application team is waiting. Where do you start?

Step 1 - Read the exact error. It tells you where to look.

PuTTY message	Meaning	First checks
Connection timed out	Packets are not reaching the server.	Server down? Network or firewall block? Wrong IP?
Connection refused	Server is reachable but SSH is not listening.	sshd stopped? SSH on another port?

PuTTY message	Meaning	First checks
Access denied / Permission denied	Reached SSH but credentials rejected.	Wrong password, locked or expired account, key problem, AllowUsers rule.
Network error: Software caused connection abort / Connection closed	Server cut the session.	Server overloaded, disk full, hosts.deny, too many sessions.

Step 2 - Quick checks from your own machine

```
ping 192.168.1.10                       # is the server alive? (may be blocked by firewall)
Test-NetConnection 192.168.1.10 -Port 22   # Windows PowerShell: is port 22 open?
telnet 192.168.1.10 22                  # shows 'SSH-2.0-OpenSSH...' if sshd answers
```

Also try: another server (is it only this one?), another user, another network, and a **jump server** (a server you can already login to): `ssh -v user@192.168.1.10` prints detailed debug lines.

Step 3 - If you still cannot login, use the console

You cannot fix a server from inside while locked out, so use an **out-of-band console**: HP iLO, Dell iDRAC, VMware vCenter console or the cloud provider's serial console. After logging in there, check:

```
systemctl status sshd                   # is the SSH service running?
ss -tlnp | grep :22                     # is it listening?
df -h                                   # a 100% full disk can block logins
uptime ; free -m                        # is the server overloaded / out of memory?
tail -50 /var/log/secure                # login errors (RHEL). Ubuntu: /var/log/auth.log
chage -l user1                          # password expired?
faillock --user user1                   # account locked after failed attempts? (RHEL 8+)
sshd -t                                 # test the SSH config for typos
firewall-cmd --list-all                 # port 22 allowed?
```

Step 4 - Escalate with evidence

If the problem is outside the server (network, firewall, security group), raise a ticket to the network/security team with what you tested: time, error message, results of ping/telnet. Good evidence saves hours.

HOW TO ANSWER IN THE INTERVIEW (say this)

'First I note the exact PuTTY error. Timeout points to network or firewall, refused points to sshd, access denied points to credentials. I test with ping and telnet on port 22, try from another jump host, and if I am still locked out I use the iLO/iDRAC or vCenter console to check sshd, disk space, load, /var/log/secure, account lock and firewall.'

Q10. I am not able to mount /data - how will you troubleshoot?

SCENARIO (a real-life situation)

After a reboot, the database team says: *'/data is empty. Our database cannot start.'* You run `mount /data` and get an error.

Simple explanation

Mounting means attaching a disk (or network share) to a folder so you can use it. A disk is useless until it is mounted - the folder `/data` is the 'door' to the disk. Permanent mounts are listed in `/etc/fstab`.

Step 1 - Read the error

Error message	Likely cause	What to do
mount point does not exist	Folder /data is missing	`mkdir /data`
special device ... does not exist	Disk not detected, or wrong device name/UUID	Check `lsblk`, `blkid`; fix fstab
wrong fs type, bad option, bad superblock	Filesystem damaged or wrong type	Check `blkid` for the type; run repair (below)
already mounted / target is busy	Something is using it	`findmnt /data`, `fuser -mv /data`
access denied by server / timed out (NFS)	Export permission or network problem	`showmount -e server`, ping, firewall

Step 2 - Is the disk visible?

```
lsblk                       # tree of disks and partitions - is the expected disk there?
blkid                       # shows each disk's UUID and filesystem type
fdisk -l                    # lists disks and partitions
pvs ; vgs ; lvs             # if LVM is used: are the volumes there?
vgchange -ay                # activate volume groups if they are inactive
```

Step 3 - Check the fstab entry

```
cat /etc/fstab              # compare the UUID/device with blkid output - look for typos
mount -a                    # try to mount everything in fstab; shows the exact error
findmnt --verify            # checks fstab for mistakes (RHEL 8+)
dmesg | tail -30            # kernel messages: I/O errors, filesystem errors
```

Step 4 - Repair if the filesystem is damaged (disk must be UNMOUNTED)

```
fsck -y /dev/sdb1           # for ext3/ext4 filesystems
xfs_repair /dev/sdb1        # for XFS filesystems (never use fsck on XFS)
mount /dev/sdb1 /data       # then try mounting manually
```

BE CAREFUL

Never run mkfs (format) on a disk that has data - it erases everything. Take a snapshot or backup before repairing. Add the option **nofail** to non-critical fstab lines and always test with `mount -a` before rebooting; a wrong fstab line can send the server to emergency mode.

HOW TO ANSWER IN THE INTERVIEW (say this)

'I read the exact mount error, confirm the disk is visible with lsblk and blkid, check LVM with pvs/vgs/lvs, verify the /etc/fstab entry and UUID, run mount -a to see the error, check dmesg for disk errors, and if the filesystem is corrupt I run fsck or xfs_repair on the unmounted device after taking a backup.'

Q11. Day-to-day activities

SCENARIO (a real-life situation)

The interviewer asks: *'Describe a normal day as a Linux admin.'* Show that you are organised: check health first, handle tickets, do planned work, and document.

A typical day

Time	Activity
Morning	Check monitoring dashboards and emails/alerts. Review overnight batch jobs and backup status. Do a basic health check on important servers.
Daytime	Work on tickets: **incidents** (something broke), **service requests** (create user, extend disk, install package), **changes** (planned work with approval).
Planned work	Patching, disk extension, upgrades, firewall/NFS changes - mostly in approved maintenance windows.
Evening	Update tickets, write handover notes for the next shift, document fixes in the knowledge base.

Main responsibilities

- **Monitoring:** CPU, memory, disk space, services, network, log errors.
- **User management:** create/delete users, reset passwords, sudo access (`useradd`, `passwd`, `chage`).
- **Storage:** extend filesystems and LVM, mount NFS shares, clean up full disks.
- **Patching and security:** apply OS updates, fix vulnerabilities found by security scans.
- **Backup and restore:** verify backups succeeded; restore files when asked.
- **Automation:** cron jobs and shell scripts for repeated tasks.
- **Troubleshooting:** slow servers, services not starting, login problems, network issues.
- **Documentation and coordination:** work with application, network, storage and security teams.

Daily health check - commands

```
uptime                      # load and how long the server has been running
df -h                       # disk space (watch anything above 80%)
free -h                     # memory
systemctl --failed          # services that failed
top                         # live view of CPU/memory users (press q to quit)
tail -50 /var/log/messages  # recent system messages (Ubuntu: /var/log/syslog)
last | head                 # recent logins
```

HOW TO ANSWER IN THE INTERVIEW (say this)

'My day starts with checking monitoring and the health of servers - load, memory, disk, failed services and backups. Then I work on incidents and service requests by priority, perform approved changes such as patching or disk extension, and finish by updating tickets and handover notes.'

Q12. Last incident / challenge you faced

BE CAREFUL

This question tests **your own** experience, so use something that really happened to you (or a practice-lab problem - say honestly that it was a lab). Interviewers can ask follow-up questions and quickly notice invented stories. Below is a **sample** to show the structure; replace the details with yours.

The STAR method - a simple way to tell the story

Letter	Meaning	Question it answers
S	Situation	What was happening?
T	Task	What was my responsibility?
A	Action	What exactly did I do, step by step?
R	Result	What was the outcome and what did we learn?

Sample story: filesystem 100% full

SCENARIO (a real-life situation)

Situation: At 2 AM we got a P1 alert - the application could not write data and users saw errors.
Task: I was on call and had to restore service quickly.

- **Action 1:** Logged in and ran `df -h` - the `/var` filesystem was 100% full.
- **Action 2:** Found the cause: `du -sh /var/* | sort -h` showed one application log of 40 GB. Log rotation was not working.
- **Action 3:** Archived the old log to another location, then emptied the file with `> app.log` (instead of rm, because the application still held the file open and rm would not free space). You can check with `lsof | grep deleted`.
- **Action 4:** Confirmed space was free, the application recovered, and users could work. Updated the incident ticket.

Result: Service was restored in about 20 minutes. Afterwards I fixed the logrotate configuration, set a monitoring alert at 80% disk usage, and wrote a root cause analysis (RCA) with the prevention steps.

HOW TO ANSWER IN THE INTERVIEW (say this)

'Describe one real incident: what broke, how you found the cause, what you did, how long it took, and what you changed so it does not happen again. Always finish with the lesson learned.'

Q13. Incident and Change management

NOTE

Your question says 'incident change'. I explain **Incident management** and **Change management** together (they are usually asked together), plus Problem and Request, because interviewers like to compare them. These come from **ITIL**, a standard way of running IT support. Tools: ServiceNow, Remedy, Jira.

SCENARIO (a real-life situation)

3 PM: The website is down. That is an **incident** - fix it now.

Next week: The manager wants to add RAM to the server on Sunday night. That is a **change** - plan it, get approval, then do it.

Type	Meaning	Example	Ticket
Incident	Unplanned interruption or slowdown of a service. Goal: restore service FAST.	Server down, disk full, app slow	INC
Problem	The root cause behind one or many incidents. Goal: find and fix the cause.	Disk full happens every month because logrotate is broken	PRB
Change	A planned modification to systems. Needs approval. Goal: change safely.	Patching, RAM upgrade, config change	CHG
Service request	A routine, low-risk request from a user.	Create a user, reset password	RITM / REQ

Incident life cycle

Detect / user reports -> Log a ticket (INC) -> Categorise and set **priority** -> Diagnose and fix (escalate if needed) -> Confirm with the user -> Close.

Priority	Meaning	Example
P1 - Critical	Business stopped; many users affected. Immediate action, 24x7, management informed.	Production server down
P2 - High	Major function broken; workaround maybe exists.	One node of a cluster down
P3 - Medium	Limited impact.	Slow response for a few users
P4 - Low	Minor issue or question.	Cosmetic problem

Each priority has an **SLA** (the agreed time to respond and resolve). After a P1, a **post-incident review / RCA** is written.

Change life cycle

- **Raise the change (CHG):** what, why, which servers, date and time window.
- **Risk and impact assessment:** who is affected? how long is the downtime?
- **Implementation plan, test plan and back-out (rollback) plan** - always have a way to undo.
- **Approval:** by the manager / **CAB** (Change Advisory Board).
- **Implement** in the approved window, **validate** with the application team, then **close** with notes.

Change type	Meaning	Example
Standard	Low risk, pre-approved, repeated task.	Add a user, extend a filesystem by a fixed size
Normal	Needs risk review and approval before work.	Monthly OS patching
Emergency	Urgent fix to restore service; fast-track approval.	Security hotfix during a P1

HOW TO ANSWER IN THE INTERVIEW (say this)

'An incident is an unplanned interruption and the goal is to restore service quickly; I log it, set priority, troubleshoot, escalate if needed and close it within SLA. A change is a planned modification that needs approval, an implementation plan and a rollback plan. A problem is the root cause behind repeated incidents.'

Q14. How will you come to know a server command?

NOTE

This question can mean two things, so I answer both: **(A)** how do you find out what a command does or which command to use when you do not remember it, and **(B)** how do you find basic information about a server.

SCENARIO (a real-life situation)

Your senior says: *'Find out how much space a folder uses.'* You forgot the command. Nobody remembers everything - good admins know how to **find** the command.

A. Finding and understanding commands

```
man du                      # manual page of a command (press q to quit, / to search)
du --help                   # short help summary
whatis du                   # one-line description
apropos "disk usage"        # search all manuals by keyword -> shows du, df, ...
which du                    # where is the command program located?
type ls                     # is it a program, alias, or built-in?
history | grep rsync        # commands I typed earlier
```

Other tricks: press **Tab** twice to auto-complete commands; press **Ctrl+R** to search your command history; and use `compgen -c` to list all commands available.

B. Finding information about the server

```
hostname                    # server name
uname -a                    # kernel version
cat /etc/os-release         # OS name and version
uptime                      # running time and load
lscpu                       # CPU details
free -h                     # memory
df -h                       # disk space
ip a                        # IP addresses
```

HOW TO ANSWER IN THE INTERVIEW (say this)

'If I do not remember a command, I use man, --help, whatis or apropos with a keyword to find it, and Tab completion and history to save time. For server details I use hostname, uname -a, /etc/os-release, uptime, lscpu, free -h and df -h.'

Q15. Patching

SCENARIO (a real-life situation)

The security team announces a critical vulnerability. Your manager says: *'All 40 Linux servers must be patched this weekend with zero surprises.'*

Simple explanation

Patching means installing updates for the operating system and software. Updates fix **security holes**, bugs and sometimes add features. Like updating your phone: necessary, but if done carelessly it can break things. That is why patching is done through a **change request**, tested first, with a backup and rollback plan.

Patching steps

Step	What you do
1. Plan	Raise a change (CHG), choose a maintenance window, inform the application owners, get approval.
2. Pre-checks	Free space in /var and /boot, repo reachable, list of running services, note the current kernel with `uname -r`, take a **VM snapshot or backup**.
3. Test first	Patch Dev/Test servers first, check the applications, then patch Production in groups.
4. Apply	Stop the application if required, install the updates.
5. Reboot	Needed when the kernel or core libraries changed.
6. Post-checks	Server up? New kernel? Services running? Disks mounted? Application team confirms.
7. Rollback if needed	Revert the snapshot, or undo with `yum history undo`, or boot the old kernel from GRUB.
8. Close	Update and close the change with notes; keep patch reports.

Commands (RHEL / CentOS)

```
uname -r ; df -h /var /boot          # BEFORE: note kernel version and free space
yum check-update                     # what updates are available?
yum update --security -y             # install only security updates (or: yum update -y)
yum update -y --exclude=kernel*      # patch everything except kernel (if needed)
needs-restarting -r                  # tells you if a reboot is required
reboot                               # restart (in the approved window only!)
uname -r ; systemctl --failed ; df -h   # AFTER: verify
yum history                          # list of transactions; undo with: yum history undo <ID>
```

Commands (Ubuntu / Debian)

```
apt update                           # refresh the list of available updates
apt list --upgradable                # see what will change
apt upgrade -y                       # install updates
ls /var/run/reboot-required          # exists = reboot needed
```

Large companies automate this with Red Hat Satellite, Ansible or Spacewalk; Windows servers use WSUS or SCCM.

BE CAREFUL

Never patch production without approval, a backup/snapshot and a rollback plan. Do not patch all servers of a cluster at the same time - patch one by one so the service stays available.

HOW TO ANSWER IN THE INTERVIEW (say this)

'Patching is installing OS updates to fix vulnerabilities and bugs. I raise a change, take a snapshot, run pre-checks, patch non-production first, then production in batches using yum update (or apt upgrade), reboot if the kernel changed, verify services, and keep a rollback plan.'

Quick revision: 30-second answers

Question	30-second answer
DNS port	53 - UDP for queries, TCP for zone transfers and large replies.
Check ports	ss -tulnp / netstat -tulnp locally; telnet, nc or nmap to test remotely.
SFTP	Secure file transfer over SSH (port 22): sftp user@host; put = upload, get = download.
CLI	Text-based interface; light, remote, scriptable. GUI = graphical. (NCL: ask what is meant.)
Booting	BIOS/UEFI + POST, GRUB, kernel + initramfs, systemd, login.
NFS / df hang	NFS server unreachable. df -hl, find mounts, ping/showmount, umount -f/-l.
Yesterday's load	sar -q -f /var/log/sa/saDD (needs sysstat); otherwise monitoring tool.
Packages	rpm -qa / yum list installed; dpkg -l / apt list --installed.
No PuTTY login	Read the error; ping/telnet 22; jump host; use iLO/iDRAC console; check sshd, disk, logs.
Cannot mount /data	Read error; lsblk/blkid; LVM; fstab; mount -a; dmesg; fsck/xfs_repair.
Day to day	Monitoring, tickets, user and disk work, patching, backups, scripts, documentation.
Incident story	STAR: Situation, Task, Action, Result + lesson learned.
Incident vs change	Incident = unplanned, restore fast. Change = planned, approved, with rollback.
Find a command	man, --help, whatis, apropos, Tab, history.
Patching	Change request, snapshot, test first, update, reboot, verify, rollback plan.

Top 25 commands for a beginner

Command	Use	Command	Use
ls / cd / pwd	List / change / show folder	df -h	Disk space
cat / less / tail -f	Read files / follow logs	du -sh *	Folder sizes
grep	Search text	free -h	Memory
find	Search files	top	Live processes
ps -ef	List processes	kill	Stop a process
systemctl status/start/stop	Manage services	journalctl	System logs
ss / netstat	Ports and connections	ping / curl	Test connectivity
useradd / passwd	Create user / password	chmod / chown	Permissions / owner
mount / umount	Attach / detach disks	lsblk / blkid	Show disks
rpm / yum / dpkg / apt	Packages	sar / uptime	Performance
scp / sftp	Copy files securely	man	Manual of a command

Interview tips for a beginner

- Be honest: *'I have not worked on that in production, but in my practice lab I would do this...'* is far better than guessing.
- Always explain **why** you run a command, not only the command name.
- Mention safety habits: backup/snapshot, change request, test first, rollback plan.
- If you do not understand a question, repeat it in your own words and ask if that is what they mean.
- Practise on a free VM (VirtualBox with CentOS/Rocky/Ubuntu) - try each command in this guide at least once.